Reading a magnet's Hamiltonian term by term: two-dimensional terahertz spectroscopy as vertex spectroscopy of $TmFeO_3$

Working draft[1]

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One-dimensional spectroscopy measures the *spectrum* of a Hamiltonian; two-dimensional coherent spectroscopy (2DCS) measures its *vertices*—the individual nonlinear coupling terms that convert one coherence into another. We show that in the rare-earth orthoferrite $TmFeO_3$ this correspondence is one-to-one and exhaustive: time-reversal parity and space-group irrep bookkeeping reduce all possible Fe–Tm conversion vertices to exactly two—a static two-magnon exchange $WS^2\lambda$ and a field-gated $\kappa^B BS\lambda$—and, with every damping rate set by a measured linewidth, every emission weight by a measured oscillator strength, and the drive given by the measured single-cycle THz waveform, one Hamiltonian reproduces the complete cross-peak inventory of all three polarization-resolved detection channels without retuning. Each observed peak identifies one vertex; each structural *absence* identifies a selection rule—most sharply, the missing $\omega_\tau = E_{13}$ rows of the same-polarization channel reveal that the $1 \to 3$ crystal-field transition is electric-dipole dark for $E \parallel c$, a polarization-resolved darkness invisible to 1D absorption, which then fixes the entire same-polarization hierarchy including its rephasing/non-rephasing structure. The resulting atlas overdetermines the microscopic model and yields six falsifiable predictions.

Introduction.—Linear response cannot distinguish two Hamiltonians with the same normal modes: absorption measures pole positions and dipole weights, and every coupling that merely hybridizes modes is absorbed into the mode basis. What linear response can never see is a *conversion*: a coherence prepared on one transition re-emitted on another. That is precisely what a cross peak in two-dimensional coherent spectroscopy (2DCS) certifies [1, 2]—and a conversion requires a specific anharmonic vertex in the Hamiltonian. A cross peak at (ω_τ, ω_t) is therefore not merely a feature to be fitted: it is a *measurement of one coupling term*, with the delay axis naming the stored coherence, the detection axis naming the emitter, and the signed side (rephasing versus non-rephasing) constraining the pathway order. If symmetry makes the space of candidate vertices finite—and in a crystal it does—then a polarization-resolved 2D atlas can be read as the Hamiltonian itself, term by term, with structural absences as informative as peaks.

Here we carry this program to completion for $TmFeO_3$, a rare-earth orthoferrite in which two qualitatively different subsystems share one THz octave: the Fe magnons and the Tm^{3+} crystal-field (CEF) transitions. Below the spin reorientation the Fe sublattice orders in Γ_2 (F_x, C_y, G_z) [3–5], with quasi-ferromagnetic and quasi-antiferromagnetic magnons at $q_{\rm FM} = 0.38$ and $q_{\rm AFM} = 0.90$ THz; the three lowest Tm CEF singlets form a qutrit with transitions $E_{12} = 0.50$, $E_{23} = 0.70$, $E_{13} = 1.20$ THz. A single-cycle THz pulse covers all five modes, and experiments in three detection channels—cross-polarized for $H \parallel a$ and $H \parallel c$, same-polarized for $H \parallel a$—observe distinct, reproducible peak inventories. We show that the entire atlas is generated by exactly two symmetry-selected Fe–Tm vertices, one magnon–magnon coupling already present in the Fe sector, and one polarization selection rule, with no free lineshape parameters.

Model and why it is exact where it matters.—The Fe^{3+} $(S = 5/2)$ sublattice is treated classically with the calibrated orthoferrite Hamiltonian (exchange, single-ion anisotropy, Dzyaloshinskii–Moriya), reproducing $q_{\rm FM}$, $q_{\rm AFM}$, and Γ_2. Each Tm^{3+} ($^3H_6$; 13 CEF singlets in C_s site symmetry) is truncated to its three lowest singlets and carried as the Gell-Mann vector $\lambda^a = \langle \Lambda^a \rangle$. Because every single-ion term is linear in the Λ^a, the classical $\vec{\lambda}$ equation of motion *is* the exact von Neumann equation of the driven, damped qutrit, and a Boltzmann ensemble over level seeds is the exact thermal density matrix [Supplemental Material (SM), Sec. S1]. The Fe–Tm sector is constructed by projecting the physical exchange $\boldsymbol{S}^T A\, \boldsymbol{J}$ into the qutrit in a transported gauge over the four Tm sublattices [SM, Secs. S2–S3].

The vertex space is small.—A candidate conversion vertex must (R1) reach the bright coordinate of the target transition, (R2) be sourced by the pumped mode, (R3) be time-reversal even, (R4) be Γ_1-allowed in the ordered state, (R5) carry the frequency mismatch (two non-λ legs minimum), and (R6) engage both pulses to survive the nonlinear subtraction $M_{\rm NL} = M_{AB} - M_A - M_B$. These six requirements exhaust the coupling space [SM, Sec. S4]. The static bilinear $K^- S\lambda$ fails threefold: linear in S, it hybridizes but cannot convert; its channels are Γ_1-cancelled to machine precision; its one allowed component reorients the ground state. The complete quadratic families W_4, W_1^{xy}, W_6, κ^E are exactly inert. Two vertices survive:

$$H_W = W_1^{yy}\, S_y^2\, \lambda^1, \qquad H_\kappa = \kappa^B_{1y}\, B_y(t)\, S_y\, \lambda^1. \qquad (1)$$

The data require both, and assign each its own peak.

All spectra below are computed with the *measured* single-cycle pump (spectrum peaked at 0.53 THz ≈

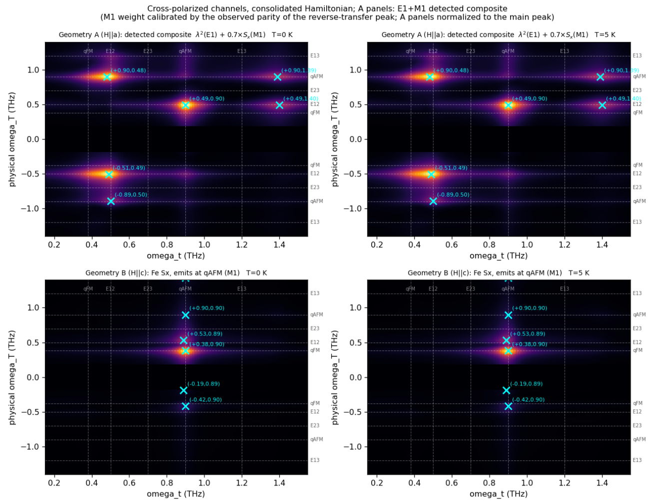

FIG. 1. Cross-polarized channels from experimental inputs only (measured pulse, linewidth damping, oscillator-strength emission). Top: geometry A ($H \parallel a$), detected E1+M1 composite (λ^2 plus M1-weighted magnon channel), normalized to the main peak: the magnetoelectric $(q_{\rm AFM}, E_{12})$ peak of W_1^{yy} and the reverse transfer $(E_{12}, q_{\rm AFM})$ of W_1^{xz} at equal intensity, as observed; the $(q_{\rm AFM}, q_{\rm AFM})$ magnon diagonal is absent, as observed, which pins the Zeeman:E1 drive ratio; parity pins the M1/E1 detection weight ($w \simeq 0.7$). Bottom: geometry B ($H \parallel c$), magnon M1 channel—the magnon–magnon $(q_{\rm FM}, q_{\rm AFM})$ peak and the κ^B-gated transfer $(E_{12}, q_{\rm AFM})$. Left $T = 0$, right $T = 5\,$K.

E_{12}, no DC), each coherence damped at its measured linewidth [$\gamma(E_{12}, E_{13}, E_{23}) = 0.038,\ 0.05,\ 0.10\,$THz Bloch; magnon Gilbert $\alpha = 1.7\times 10^{-3}$], and each emitter weighted by its measured oscillator strength ($\sqrt{f} = 2.5$–3.0 for the E1-bright CEF lines; $f_{q_{\rm AFM}} = 7 \times 10^{-4}$: the $q_{\rm AFM}$ magnon is E1-dark but M1-bright) [SM, Sec. S5]. Peaks are identified blindly as 2D local maxima with prominence ≥ 1.5 over the local median background. The E1/M1 dichotomy already dictates the readout physics: a magnon can only be seen either by borrowing a CEF electric dipole through an Fe–Tm vertex, or through its own magnetic dipole at the magnon frequency. The three channels realize the three possibilities.

Peak 1, geometry A cross-polarized: the two-magnon vertex.—During the delay the Zeeman-launched $q_{\rm AFM}$ magnon precesses; the static vertex W_1^{yy}—the term by which a magnon distortion modulates the CEF splitting—rectifies the two-pulse cross term of S_y^2 into a step force on the bright E_{12} coordinate,

$$M_{\rm NL}^{W} \propto W_1^{yy} A^2 \cos(q_{\rm AFM}\tau)\,[1 - \cos E_{12} t], \qquad (2)$$

a factorized, rank-1 cross peak at $(q_{\rm AFM}, E_{12})$ [Fig. 1, top]. The E1-dark magnon borrows the Tm dipole and emits at a CEF frequency: the peak is a magnetoelectric object, and its existence measures W_1^{yy} directly. The E_{12} quantum is drawn from the pulse-edge bandwidth (an intra-pulse difference-frequency process): stretching the pulse from $\sigma = 0.25$ to 1.0 at fixed amplitude collapses the peak by 4.5 decades—the adiabatic suppression a displacive mechanism demands—while switching off the direct Tm drive changes it by $< 10^{-4}$ [SM, Sec. S6]. The same channel also carries the *reverse* transfer: a weaker genuine peak at $(E_{12}, q_{\rm AFM}) \approx (0.5,\ 0.9$–$1.0)$, the stored E_{12} coherence converted *into* the magnon by the static hybridization W_1^{xz} (it dies sevenfold when $W_1^{xz} \to 0$ and scales linearly with the Tm drive; a $2E_{12} = 1.0\,$THz harmonic companion sits beside it). The static W family is thus read in both directions—magnon→CEF through W_1^{yy}, CEF→magnon through W_1^{xz}. Because the reverse peak emits at the magnon frequency, it is detected through the magnon's *own* M1 dipole, whereas the forward peak uses the Tm E1 dipole. Two observed facts then fix two detection-side dials with no Hamiltonian change: the *absence* of the $(q_{\rm AFM}, q_{\rm AFM})$ magnon diagonal pins the Zeeman drive far below the E1 drive (the diagonal is cubic in the Zeeman amplitude, the main peak quadratic, the reverse peak linear), and the intensity *parity* of the two transfer peaks calibrates the M1/E1 detection weight ($w \simeq 0.7$ at the resulting operating point). The $E_{12}{+}q_{\rm AFM} = 1.4\,$THz combination tones ($\sim 0.4$) remain as predicted companions [SM, Sec. S7].

Peaks 2–3, geometry B cross-polarized: the gated vertex, and a division of labor.—For $H \parallel c$ both observed peaks emit at $q_{\rm AFM}$ through the magnon M1 dipole. $(q_{\rm FM}, q_{\rm AFM})$ is a pure magnon–magnon conversion (anisotropy plus DM couple the branches): it measures the Fe sector alone. $(E_{12}, q_{\rm AFM})$ isolates the second Fe–Tm vertex: with κ_{1y}^B alone it is present; with the static hybridization alone it is absent, and no strength rescues it—a sweep shows the static vertex only merges and shifts the two delay rows (a mixer moves peaks; a gated converter adds them) [SM, Sec. S7]. The gate $B_z(t)$ acts only during the pulse, impulsively imprinting the stored E_{12} phase on the magnon—and the same gate makes κ^B *identically silent* in geometry A ($B_z = 0$), where its background would contaminate the clean W peak. The observed inventory—$(q_{\rm FM}, q_{\rm AFM}) = 1.00$, $(E_{12}, q_{\rm AFM}) = 0.46$, residuals ≤ 0.17, CEF channels exactly zero by subalgebra closure—requires the gated vertex. The reported post-drive *buildup* of the geometry-B signal, by contrast, is reproduced by *no* coherent vertex once the measured linewidths are enforced: every coherence reservoir dies within its T_2 (E_{12}: 8 ps), so the converted magnon amplitude peaks at pulse end and decays with the reservoir lifetime. A slow buildup therefore implicates the one long-lived reservoir the ion possesses—the CEF *populations*—fed incoherently by the dissipated pulse energy and acting on the magnon through the population channel (λ^3, W_3) of the same exchange: the mechanism of thermally driven spin reorientation. Implemented in the solver (dissipated coherence energy shifts the λ^3 equilibrium; the relaxation chases it), this channel reproduces the observation—the transfer signal falls off the impulsive peak, recovers on the repopulation time $1/\Gamma_3$, and persists beyond 100 ps—and its recovery time and late-time fraction directly measure the Tm repopulation time and $c_{\rm heat}W_3$ [SM, Sec. S7].

Peaks 4–8, geometry A same-polarized: an absence becomes a selection rule.—The same-polarization channel ($E \parallel c$ in and out) shows five features, ordered $(E_{12}, E_{23}) \approx (E_{12}, 0) > (E_{12}, E_{13}) > (q_{\rm AFM}, 0.38), (q_{\rm AFM}, E_{13})$, with *both* CEF transfer peaks on the non-rephasing side. The decisive information is what is missing: no feature anywhere carries the delay phase $E_{13} = 1.2\,$THz. A bright $1 \to 3$ dipole would store an E_{13} delay coherence at first order and produce an (E_{13}, E_{23}) peak with *exactly* the same dipole product $\mu_{12}\mu_{13}\mu_{23}$ as the observed (E_{12}, E_{23}) peak—no parameter can separate them. The absence of the whole row is therefore a selection rule,

$$\mu_{13}^z \simeq 0, \tag{3}$$

the natural consequence of a parity-alternating CEF ladder ($|1\rangle, |3\rangle$ even, $|2\rangle$ odd under the site mirror). The measured E_{13} oscillator strength ($f = 8.7$) lives in the orthogonal polarizations: the same transition is bright in one polarization and dark in the other, a fact the 2D map resolves and 1D absorption, summed over its bright polarizations, does not.

Equation (3) then fixes the entire channel [Fig. 2]. With the single-action (rephasing) route $\rho_{12} \to \rho_{32}$ removed, the dominant transfer runs coherence→population→coherence—one probe action converts the stored E_{12} coherence into a level-2 population still carrying the delay phase, a second converts it into the emitting E_{23} coherence—a co-rotating route that lands *non-rephasing*, the observed side. Its zero-frequency output leg is the $(E_{12}, 0)$ peak, a sharp E_{12} line in the rectified signal $S_{\rm DC}(\tau)$. The partner (E_{12}, E_{13}), though lower order, emits through the throttled μ_{13}^z and is capped at 0.78 of the dominant peak; the measured ratio calibrates the residual admixture at the percent level. The magnon-delay peak $(q_{\rm AFM}, E_{13})$ is weak and thermally activated ($0.10 \to 0.51$ from 0 to 20 K). The one feature the model does not produce is $(q_{\rm AFM}, 0.38)$: the $q_{\rm FM}$ emission carrier exists (a b-axis M1 ridge, as Γ_2 requires) but no $q_{\rm AFM} \to q_{\rm FM}$ transfer vertex of sufficient strength exists in the present Hamiltonian class—itself a sharp statement about what the material's Hamiltonian must contain beyond it [SM, Sec. S8].

Because the single-ion dynamics is quantum-exact, these pathway claims were verified against a standalone qutrit integrator (same pulse, same dampings), which reproduces the lattice census exactly and establishes two robust residuals as predictions: a rephasing remnant at $(-E_{12}, E_{23})$ at ~ 0.5 of the dominant peak (an exact single-ion floor under this pulse) and a lattice two-quantum line at $\omega_\tau = 2E_{12} = 0.99\,$THz, absent in the single-ion dynamics and hence a pure inter-site product [SM, Sec. S8].

The atlas as an overdetermined measurement.—The ledger is now complete, and it is worth stating in the vertex-spectroscopy language:

observation	Hamiltonian content it measures
$(q_{\rm AFM}, E_{12})$, geom. A	two-magnon vertex W_1^{yy}
pulse-width collapse	. . . and its displacive mechanism
$(E_{12}, q_{\rm AFM})$, geom. B	gated vertex κ_{1y}^B
its silence in geom. A	. . . and its B_z gate
$(E_{12}, q_{\rm AFM})$, geom. A	reverse transfer: W_1^{xz}
post-drive buildup	population (T_1) sector [SM, S7]
$(q_{\rm FM}, q_{\rm AFM})$, geom. B	Fe anisotropy/DM sector
missing $\omega_\tau{=}E_{13}$ rows	selection rule $\mu_{13}^z \simeq 0$
NR side of both transfers	. . . and the pathway order it forces
$(E_{12}, E_{13})/(E_{12}, E_{23})$	residual μ_{13}^z admixture
$(q_{\rm AFM}, 0.38)$ unexplained	a coupling the model lacks

Eight independent observations constrain four microscopic quantities and one selection rule: the map overdetermines the model, which is what makes it falsifiable. Concretely: (1) any peak emitting at $q_{\rm AFM}$ inherits the $15\times$ sharper magnon linewidth along ω_t; (2) $(q_{\rm FM}, q_{\rm AFM})$ is temperature-flat while $(E_{12}, q_{\rm AFM})$ falls as $n_1 - n_2$ and $(q_{\rm AFM}, E_{13})$ grows steeply; (3) rotating either polarization away from c revives the $\omega_\tau = 1.2\,$THz rows; (4) lowering the fluence inverts the (E_{12}, E_{23}) versus (E_{12}, E_{13}) order (they differ by one field power); (5) the rephasing remnant at $(-E_{12}, E_{23})$ and the two-quantum line at $\omega_\tau = 0.99\,$THz (resolvable from 0.90) should exist in raw data; (6) adiabatically long pulses collapse the geometry-A cross peak.

Conclusion.—Two-dimensional THz spectroscopy of $TmFeO_3$ admits a term-by-term microscopic reading: two symmetry-selected Fe–Tm vertices, one Fe-sector magnon–magnon coupling, and one polarization-resolved selection rule generate the complete measured atlas, computed from measured inputs with no free lineshape parameters. The program—bound the vertex space by symmetry, assign each peak and each absence, and let the atlas overdetermine the model—is not specific to $TmFeO_3$: it applies to any magnet whose 2D map spans distinct subsystems, and it elevates 2DCS from a spectroscopy of modes to a spectroscopy of Hamiltonian terms.

Derivations, numerical protocol, and the complete falsification ledger are given in the Supplemental Material; the long-form working note and all run recipes are archived with the repository.

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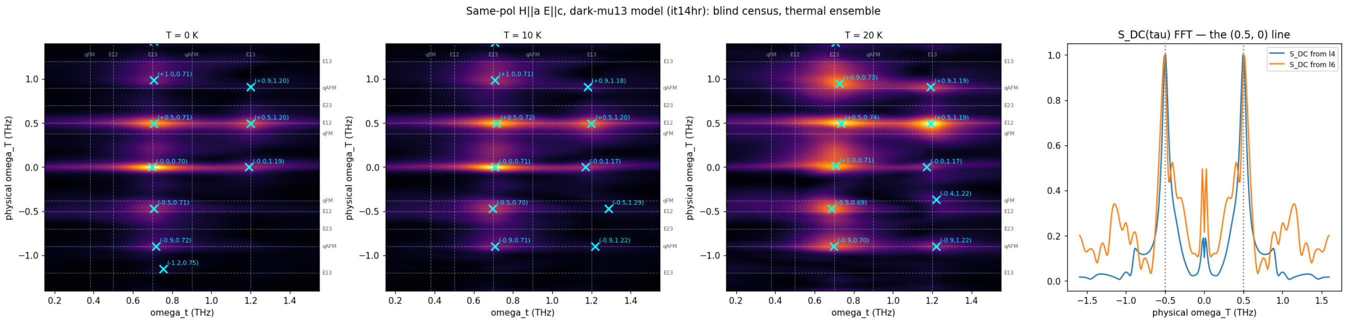

FIG. 2. Same-polarized geometry-A channel (blind census at 0, 10, 20 K; crosses mark genuine local maxima). Both transfer peaks, $(+E_{12}, E_{23})$ and $(+E_{12}, E_{13})$, are non-rephasing with the E_{23} member dominant—the fingerprint of the z-dark $1 \to 3$ dipole. The rephasing remnant at $(-E_{12}, E_{23})$ and the two-quantum line at $\omega_\tau = 2E_{12}$ are predictions. Right: the rectified signal $S_{\mathrm{DC}}(\tau)$, whose sharp line at $+E_{12}$ is the $(E_{12}, 0)$ peak.

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